

**MEDIATEK**

# MT8676 Android DDR User Manual

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Version History

| Version | Date       | Author        | Description      |
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# 1 DDR

## 1.1 Overview

### 1.1.1 Brief introduction

This chapter introduces the MT8676 DRAM configuration, and the DRAM verification process (CVI flow).

### 1.1.2 Abbreviations

**Table 1-1. Abbreviations**

| Abbreviation | Explanation                   |
|--------------|-------------------------------|
| CVI          | Customer Verified Item        |
| DDR4         | Double Data Rate 4            |
| DRAM         | Dynamic Random Access Memory  |
| HDK          | Hardware Design Kit           |
| LPDDR4X      | Low-Power Double Data Rate 4X |
| LPDDR5X      | Low-Power Double Data Rate 5X |
| MOL          | MediaTek Online               |
| MTK          | MediaTek                      |
| PCB          | Printed Circuit Board         |
| QVL          | Qualified Vendor List         |

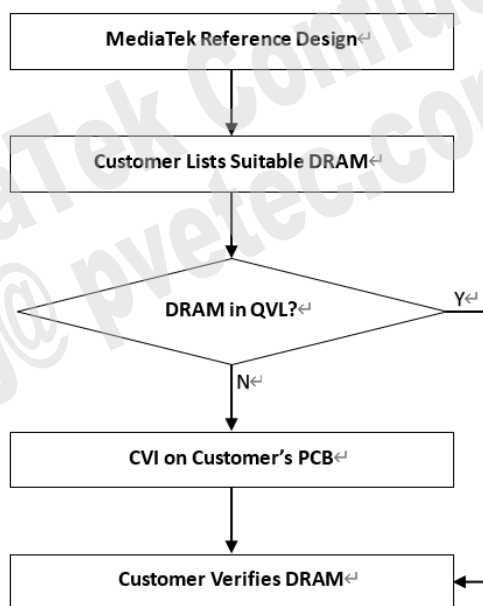
## 1.2 Architecture/Process Overview

### 1.2.1 Customization Process

After MediaTek releases the Hardware Design Kit (HDK) for reference, customers can choose appropriate DRAM projects based on self needs.

MediaTek's Qualified Vendor List (QVL) is the DRAM material that MediaTek has passed DRAM verification on the MediaTek reference board during the IC verification stage. If the DRAM material selected by the customer is not in QVL, the customer will need to verify the new DRAM project on the client's PCB. New DRAM projects completed by customer verification will be added to the MediaTek Customer Verified Items (CVI) list.

The details are shown as [Figure 1-1](#):



**Figure 1-1. DRAM customization process**

## 1.2.2 Process for Adding Item to CVI

### 1.2.2.1 New DRAM Verification - CVI flow

Figure 1-2 describes the process of how customers submit new DRAM materials for verification and add them to CVI.

Please contact MediaTek's CPM first to obtain permission to add new DRAM projects to CVI firstly, then perform the DRAM stress test according to the DRAM verification SOP, and send the test results to MediaTek, MediaTek will review the verification results of the new DRAM project, if the results is okay, the new DRAM project will be added to the MediaTek CVI list.

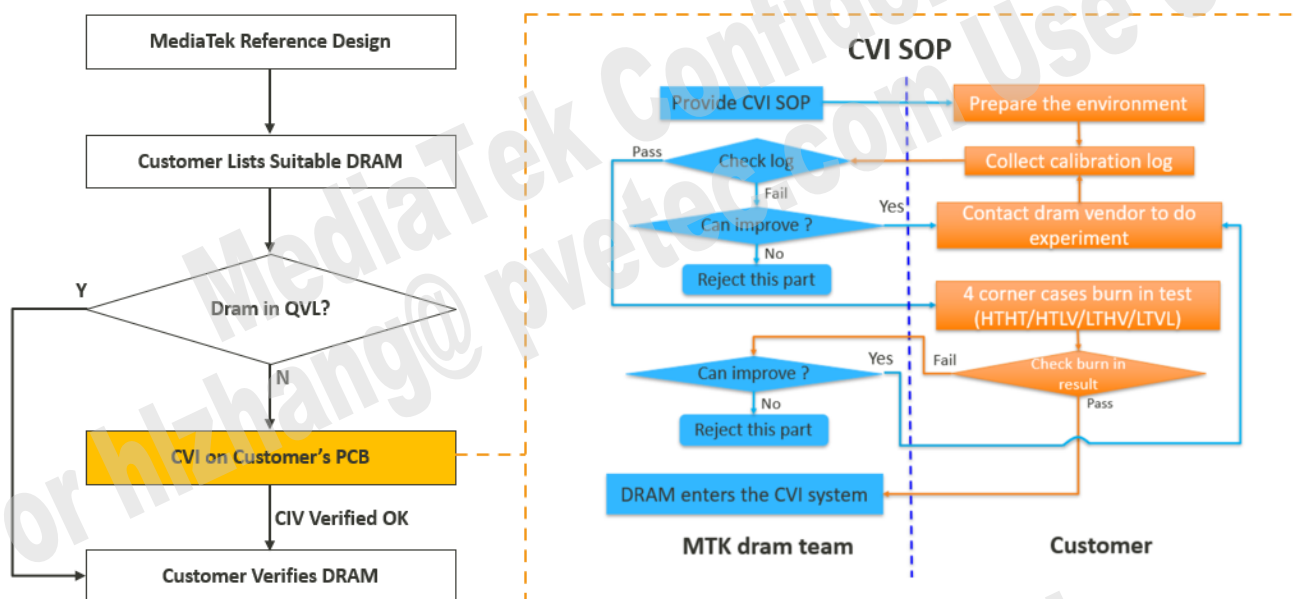


Figure 1-2. Process for adding item to CVI

### 1.2.2.2 CVI Preparation

Customers can check whether the DDR to be used is in the list at <https://online.mediatek.com/apps/qvl/>. If it is in the list and the customer has followed the public version of MMD (the hardware design is consistent with the MTK public version), the customer can directly use this DDR.

If the DDR is not in the QVL list, the customer submits the requirement to the PM/BM. After the PM/BM agrees, the customer submits a CR, attaches the DDR Spec and provides the following information:

Customer === Customer's name

Chip === MT8676 + DDR part number

DRAM PN === DRAM Part Number

DRAM Die === B die

Provide dram date sheet === (Necessary)

Provide dram type === e.g: LPDDR5X

Provide system dram speed === The maximum frequency that can support

Provide PCB layout === (Necessary)

Whether MMD or not? === YES

Have you done SI/PI simulation? === (Necessary)

Whether it is applied according to the SOC spec (over clocking, changing 2CH to 1CH...) === It is applied according to SOC spec and does not over clock.

HW CPM === MTK HW CPM

SW CPM === MTK SW CPM

BM === MTK BM

### 1.3 Configuration/Customization Guideline

#### 1.3.1 DRAM Type

The DRAM type supported by MT8676 is LPDDR5X, which supports single rank and dual rank DRAMs. The highest DRAM data rate supported by MT8676 is 7500 Mbps.

#### 1.3.2 DRAM Size

The maximum DRAM size supported is 24 GB.

MediaTek implements a DRAM size adaptation mechanism. If new DRAMs of different sizes are used, customers do not need to configure the DRAM size separately or update the memory size in the Kernel device tree.

#### 1.3.3 DRAM Power Voltage

The typical voltages of dram are described in [Table 1-2](#).

**Table 1-2. MT8676 DRAM power voltage**

| Type              | Vcore | VDD1 | VDD2H | VDD2L | VDDQ | VMDDR |
|-------------------|-------|------|-------|-------|------|-------|
| Voltage (Unit: V) | 0.8   | 1.8  | 1.05  | 0.91  | 0.5  | 0.85  |

#### 1.3.4 DRAM Customization Guideline

1. Update MemoryDeviceList\_MT6897.xls to support new DRAM projects

MediaTek will provide MemoryDeviceList\_MT6897.xls for new DRAM projects

[Figure 1-3](#) demonstrates support for Micron MT62F2G32D4DS\_023

| Vendor  | Part Number       | Type             | Density (Mb) | Board ID   | NAND/eMMC ID | MT6895  |            |
|---------|-------------------|------------------|--------------|------------|--------------|---------|------------|
|         |                   |                  |              |            |              | CHIP_ID | MODE_REG5  |
| Hynix   | H58G66BK8QX067    | Discrete LPDDR5X | 65536+65536  | MT8676_EVB |              | mt6897  | 0x00000006 |
| Hynix   | H58GG8AK8BX103    | Discrete LPDDR5X | 98304+98304  | MT8676_EVB |              | mt6897  | 0x00000006 |
| Micron  | MT62F2G32D4DS_023 | Discrete LPDDR5X | 65536+65536  | MT8676_EVB |              | mt6897  | 0x000000FF |
| Micron  | MT62F3G32D8DV_023 | Discrete LPDDR5X | 98304+98304  | MT8676_EVB |              | mt6897  | 0x000000FF |
| Samsung | K3KL9L90DM_MFHCU  | Discrete LPDDR5X | 65536+65536  | MT8676_EVB |              | mt6897  | 0x00000001 |

**Figure 1-3. MT8676 MemoryDeviceList\_MT6897.xls**

**Note:**

- **Part number** and **Board ID** must be consistent with custom\_MemoryDevice.h

\vendor\mediatek\proprietary\bootable\bootloader\preloader\custom\CUSTOM\_PROJECT\inc\custom\_MemoryDevice.h

2. Update custom\_MemoryDevice.h to support new DRAM projects

```

#define BOARD_ID MT8676_EVB

#define CS_PART_NUMBER[0] H58G66BK8QX067
#define CS_PART_NUMBER[1] H58GG8AK8EX103
#define CS_PART_NUMBER[2] MT62F3G32D4DS_023
#define CS_PART_NUMBER[3] MT62F3G32D8DV_023

#endif /* __CUSTOM_MEMORYDEVICE__ */

```

Figure 1-4. custom\_MemoryDevice.h

## 1.4 Frequently Asked Questions/Troubleshooting

### 1.4.1 Enable Dram Debug Log

The dram debug log is not enabled by default. Enabling the DRAM debug log can check DRAM calibration results, DRAM self-test and other information, such as:

```

\vendor\mediatek\proprietary\bootable\bootloader\preloader\platform\mt6897\src\drivers\inc\dramc_common.h
@@ -249,7 +249,7 @@
#define mcSHOW_ERR_MSG(_x_) { mcPRINTF(_x_, ##__VA_ARGS__); }
#define mcDUMP_REG_MSG(_x_)
#else
- #define mcSHOW_DBG_MSG(_x_)
+ #define mcSHOW_DBG_MSG(_x_) { mcPRINTF(_x_, ##__VA_ARGS__); } /* Enable dram debug log */
#define mcSHOW_DBG_MSG2(_x_)
#define mcSHOW_DBG_MSG3(_x_)
#define mcSHOW_DBG_MSG4(_x_)

```

Customers usually only need to enable "mcSHOW\_DBG\_MSG(\_x\_)" to confirm dram calibration results or CVI review.

### 1.4.2 Dram Calibration Results

Customers can search the keyword "Calibration Summary" in the bootup log to view the DRAM calibration results.

Figure 1-5 demonstrates the calibration results at frequency 1600 MHz (data rate 3200 Mbps).

```

[Calibration Summary] Frequency 1600
CH 0, Rank 0
SW Impedance      : PASS
8 Phase           : FAST K
DUTY Scan         : FAST K
ZQ Calibration     : PASS
Jitter Meter      : NO K
CBT Training      : NO K
Write leveling(PI) : FAST K
Write leveling(DLY) : NO K
DUTY CYCLE MONITOR : NO K
RX DQS gating     : FAST K
RX OFFC           : NO K
RX DQ/DQS(RDDQC)  : FAST K
RX DQ/DQS(RDDQC_DQM_ONLY) : FAST K
TX DQ/DQS         : PASS
RX DATLAT         : FAST K
RX DQ/DQS(Engine) : FAST K
RX RDQS DCA       : NO K
RX RDQS DCC       : NO K
TX OE             : NO K
All Pass.

```

Figure 1-5. Dram calibration summary

### 1.4.3 Memory Test Results

After completing the DRAM calibration, a memory test will be executed to check whether the basic read and write operations of the DRAM can operate normally.

Figure 1-6 demonstrates the test results of memory test:

```

[MEM] 1st complex R/W mem test pass (start addr:0x50700000)

```

Figure 1-6. Memory test results

### 1.4.4 Command to Get Dram Information

#### 1. Dram Data Rate

The following is the command to view DRAM data rate

UART (or ADB shell) window:

```
cat /sys/bus/platform/drivers/dramc_drv/dram_data_rate
```

The result is as following:

```

console:/ #
console:/ # cat /sys/bus/platform/drivers/dramc_drv/dram_data_rate
DRAM data rate = 1600
console:/ #

```

## 2. Memory Size

The following is the command to view Memory Size:

UART (or ADB shell) window:

```
cat /proc/meminfo
```

The result is as following:

```
console:/ #  
console:/ # cat /proc/meminfo  
MemTotal:      15382408 kB  
MemFree:       11883900 kB  
MemAvailable:  13583276 kB
```

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